

Thm. Let f be analytic in the disk $|z - z_0| < R$,
 and $\{z_n\}$ be a sequence of distinct points converging to z_0 .
 If $f(z_n) = 0$ for all $n \in \mathbb{N}$, then $f \equiv 0$ in $|z - z_0| < R$.

Proof. Since f is analytic, the Taylor expansion

$$f(z) = a_0 + \sum_{k=1}^{\infty} a_k (z - z_0)^k$$

is valid in $|z - z_0| < R$. Since f is continuous,

$$\lim_{n \rightarrow \infty} f(z_n) = f(z_0) = a_0.$$

so $a_0 = 0$. So, write

$$f(z) = (z - z_0) \cdot \left(a_1 + \sum_{k=2}^{\infty} a_k (z - z_0)^{k-1} \right)$$

let $z = z_n$, so

$$\lim_{n \rightarrow \infty} \frac{f(z_n)}{z_n - z_0} = a_1 = 0, \quad \text{inductively, } a_0 = a_1 = \dots = 0.$$

Corollary. If f is analytic at z_0 , then there is $r > 0$ such that

$$f(z) \neq 0 \quad \text{for all } 0 < |z - z_0| \leq r$$

or $f \equiv 0$

Proof. If no such $r > 0$ exists, then for each $n \in \mathbb{N}$, there is $z_n \in B_{1/n}^c(z_0)$ s.t.
 $f(z_n) = 0$.

Clearly $z_n \rightarrow z_0$, so $f \equiv 0$.

The Uniqueness Theorem.

Let f be analytic in a domain D , and that $\{z_n\}$ is a sequence of distinct points, converging $z_0 \in D$.

If $f(z_n) = 0$ for all $n \in \mathbb{N}$, then $f \equiv 0$ throughout D .

Proof. Define two sets:

$$A := \{a \in D \mid f \equiv 0 \text{ on } B_r(a) \text{ for some } r > 0\}$$

$$B := \{a \in D \mid f \neq 0 \text{ on } B_r(a) \setminus \{a\} \text{ for some } r > 0\}$$

[Lemma. Let f be an analytic in the $B_R(z_0)$.

If the set $Z(f) = f^{-1}[0]$ has a limit point in $B_R(z_0)$, then $f \equiv 0$ on $B_R(z_0)$.

Proof. Let $z_0 \in B_R(z_0)$ be the limit point of $Z(f)$, by the hypothesis. Since f is analytic, the representation

$$f(z) = a_0 + \sum_{k=1}^{\infty} a_k (z - z_0)^k$$

is valid, and since z_0 is the limit point of $Z(f)$, there is a sequence $\{z_n\}$ in $Z(f)$, converging to z_0 . Therefore,

$$\lim_{n \rightarrow \infty} f(z_n) = f(z_0) = a_0$$

is zero. Then, now,

$$f(z) = \sum_{k=1}^{\infty} a_k (z - z_0)^k = (z - z_0) \cdot \left(a_1 + \sum_{k=2}^{\infty} a_k (z - z_0)^{k-1} \right)$$

And, similarly, we can construct $\{z_n\}$ not containing z_0 , so that

$$\frac{f(z_n)}{z_n - z_0} \rightarrow 0 \quad , \text{ so } a_1 \text{ is also } 0. \text{ Inductively, we obtain } a_n = 0 \text{ for all } n \in \mathbb{N}. \quad \square$$

[Corollary. If f is analytic at z_0 , then either

$f \equiv 0$ in some $B_r(z_0)$ or $f \neq 0$ in some $B_{r'}(z_0) \setminus \{z_0\}$.

Proof. If there is no such r' exists, that is, for all $r' > 0$, there is $z \in B_{r'}(z_0)$ s.t. $f(z) = 0$, then $f \equiv 0$ for some neighbor, by the Lemma above.

[Uniqueness Theorem.

Let f be an analytic in a domain D , and $Z(f)$ has a limit point in D .

Then $f \equiv 0$ on D .